

Haigang Zhou

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EDUCATION

Ph.D. Student in Computer Science and Technology

Sep. 2023 – Present

Nanjing University, Advisor: Prof. Penghui Yao

B.S. in Mathematics and Applied Mathematics

Sep. 2019 – June 2023

Tsinghua University

PROFESSIONAL EXPERIENCE

Beijing Yongyuan Technology Co., Ltd. (Startup)

May 2026 – Present

Power Quantitative Trading Algorithm Engineer; Founding Team Member

- Developed spread prediction models for spot electricity markets in Fujian, Zhejiang, Mengxi, and other regions, alongside customer-side and generation-side power forecasting models, using models such as LightGBM, XGBoost, TCN, and Transformer.
- Generated electricity-trading submission plans for clients and moved them into live-market testing; spot-market arbitrage backtests achieved an **84.1%** daily win rate and **79.7%** weighted tick-level accuracy.
- Achieved a team ranking of **26/6,342 (top 0.41%)** in the Electricity Market Trading Track of the 4th World AI4S Prize.

MiniMax

Apr. 2025 – July 2025

LLM & Agent Intern

- Contributed to SFT/RL data iterations for M1, M2, and Xingye, focusing on building high-quality Math data; independently owned data collection, cleaning, decontamination, and quality filtering, and evaluated data impact through training runs. MiniMax-M1-80k achieved 86.0% accuracy on AIME 2024.
- Participated in the cold-start stage of the M2 Agent model, working on data construction and training-effect validation.
- Invited as a company representative to attend the 2025 WAIC forum “Mathematical Boundaries and Foundational Reconstruction of Artificial Intelligence.”

Beijing Huifeng Technology Co., Ltd. (PINNIX)

Dec. 2023 – Mar. 2024

Large Language Model (LLM) Intern

- Designed and implemented an LLM-driven, iterative question-bank workflow built around a **solve–generalize–summarize** cycle, delivering approximately 3,000 Math and Reasoning problems and continually expanding the question bank to provide core data and AI capabilities for the company’s talent assessment system.

Learning and Perception (LEAP) Lab

Jan. 2022 – May 2022

Department of Automation, Tsinghua University

Reinforcement Learning Research, Supervisor: Assoc. Prof. Gao Huang

- Investigated self-distillation for offline RL and analyzed experimental results on MuJoCo.
- Contributed to research and theoretical analysis of Meta-Weight networks that learn weights for offline RL updates.

PUBLICATIONS

Dimension-Free Approximate Tensorization of Quantum Hypercontractivity for Qudit

2026

Depolarizing Semigroups

*Yangjing Dong, Li Gao, Fengning Ou, Penghui Yao, **Haigang Zhou***

Authors listed alphabetically by surname

AQIS 2026; IEEE TIT (under review); arXiv:2606.17729 [quant-ph]

MANUSCRIPTS UNDER REVIEW

Fooling Thresholds of Halfspaces

2026

*Minglong Qin, Penghui Yao, Mingnan Zhao, **Haigang Zhou***

Authors listed alphabetically by surname

SODA 2027 (under review); arXiv:2609.21329 [cs.CC]

PROJECTS

Japanese Study Studio (jp-study-site)

Sep. 2026

Personal Project

Developed a local AI Japanese learning site in Python and JavaScript, integrating Codex CLI and Anki’s native review scheduling through AnkiConnect. Implemented interactive tutoring, persistent cross-session memory, and structured storage for learning records, textbooks, and supplementary reading materials.

TEACHING EXPERIENCE

Computational Complexity, Nanjing University

Spring 2026

Teaching Assistant, Teacher: Prof. Penghui Yao

Foundations of Data Science, Nanjing University

Fall 2025

Teaching Assistant, Instructor: Assoc. Prof. Mingmou Liu

Quantum Computing Talent Development Program, USTC

Summer 2025

Mentor, Program Lead: Assoc. Researcher Zhaofeng Su

Information Theory, Nanjing University

Spring 2025, Spring 2024

Teaching Assistant, Teacher: Prof. Penghui Yao

ACHIEVEMENTS

Outstanding Graduate Student, Nanjing University

Dec. 2025

President’s Scholarship, Nanjing University

Oct. 2023

Research Innovation Award, Department of Mathematical Sciences, Tsinghua University

Mar. 2023

Second Prize, Chinese Mathematics Competitions for College Students (Major Group)

Mar. 2020

Silver Prize, CMO (Chinese Mathematical Olympiad Winter Camp)

Dec. 2018

LANGUAGES

Computer Languages

Python (& Pandas), C/C++, Golang, Matlab, L^AT_EX.

Foreign Languages

English: CET-4 (594/710), CET-6 (616/710)
French: TFU-4, Excellent (90.5/100),
German: PHD-4, Excellent (94/100)